

ADRIJA CHAKRABORTY

OBJECTIVE

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Actively seeking full-time opportunities in Circuit Design, Physical Design and ASIC related roles.

EDUCATION

• Purdue University

West Lafayette, IN, USA

Master of Science in Electrical and Computer Engineering - VLSI and Circuit Design (Thesis Track)

Aug 2024 - May 2026

- Relevant Coursework: MOS VLSI Design, Digital Systems Design Automation, Semiconductor Memory Technologies

• Vellore Institute of Technology

Vellore, Tamil Nadu, India

Bachelor of Technology in Electronics and Instrumentation Engineering

Aug 2018 - June 2022

SKILLS

- **Programming Languages:** SPICE, Verilog, Python, Java, C, C++
- **Tools:** Cadence Virtuoso, ModelSim, Xilinx Vivado, Quartus Prime, MATLAB

EXPERIENCE

• Qualcomm (NUVIA Custom Circuits - CPU Memory Design Team)

May, 2025 - Aug, 2025

Interim Engineering Intern

- Designed a Low-power Flying-Bitline Architecture for a 6T SRAM-based macro using a single-ended read scheme without a Sense Amplifier.
- Conducted detailed HSPICE simulations to characterize key performance metrics including dynamic power consumption and read access latency.
- Reduced the bitline and senseline capacitances by 15 % and 43 % respectively, leading to an overall power improvement of around 26 % for the array and IO units of the macro.

• SAP Labs India

July, 2022 - July, 2024

Developer Associate

- Developed and optimized various backend services for SAP Cloud for Customer and SAP ByDesign products.

• Samsung

July, 2020 - March, 2021

Research Internship

- Implemented a framework in Python to separate speech and music from mixed audio, enhancing audio source separation accuracy by 25.37 %.

ACADEMIC PROJECTS

• Mitigating Non-Idealities in Analog CiM Systems, Advisor - Dr. Sumeet Gupta

Aug 2024 - Present

Python, HSPICE

- Developed a framework with circuit-level techniques to mitigate non-ideal effects in Compute-in-Memory (CiM) based DNN accelerators implemented using 8T SRAM crossbar arrays at the 7nm technology node.
- Achieved near-ideal software accuracy for DNN accelerators and upto 76.06% accuracy gain for BNN accelerators.

• Design of 8-bit Wallace Tree multiplier with Multiply-Accumulate (MAC) module

Sept 2024 - Dec 2024

Cadence Virtuoso



- Designed the schematic circuit for an 8-bit Wallace Tree Multiplier using 4:2 compressors and achieved a layout area of $1916.57 \mu m^2$, worst-case propagation delay of $1.42 ns$ and energy consumption of $609.038 fJ$ at 1.1 V supply.
- Developed a pipelined Multiply-Accumulate (MAC) module using this multiplier and implemented its layout with an area of $2954.77 \mu m^2$, a clock-to-Q delay of $850.85 ps$ and energy consumption of $1.29 pJ$ per MAC operation.

• Energy Efficient Approximate Multiplier for Image Processing (Technical Publication)

Jan 2022 - May 2022

ModelSim, MATLAB



- Designed an 8-bit approximate Dadda Multiplier in Verilog, incorporating two imprecise 4:2 compressors and an input-reordering circuit.
- Reduced Power Delay Product (PDP) by 20.19 % and achieved 97.59 % Mean Structural Similarity Index Measure (MSSIM) for various image processing applications.

HONORS AND AWARDS

• Winner of Apple Design Contest

2024

Purdue University

- Secured first place in the design contest organized by Apple for the Wallace Tree Multiplier project.

GRADUATE ASSISTANTSHIP

• Graduate Teaching Assistant for MOS VLSI Design (ECE 55900)

Aug 2025 - Dec 2025

Purdue University

- Conducted Cadence Virtuoso sessions for students and assisted them with assignments in digital circuit design.